Next Steps in Multi-Agent Systems

Presenter:
Deborah Dahl
Conversational Technologies, Open Voice Interoperability Initiative

Contributors:
David Attwater
Csaba Bolyós
Emmett Coin
Diego Gosmar
Andreas Zettl

Open Source Summit North America
May 18-20, 2026
Minneapolis, Minnesota, USA

OPEN VOICE INTEROPERABILITY

Outline

- Why multi-agent systems?
- Multi-agent use cases
- Open Floor Protocol (OFP) and Agent to Agent (A2A)
- OFP examples
- Future directions in multi-agent systems
- Getting involved in OFP

Why Multi-Agent Systems?

Similar arguments to microservices or Object-Oriented Programming

- Separation of concerns
- Encapsulation of capabilities
- Maintainability
- Testing
- Reuse of specialized agents
- Leverage third-party agent ecosystem
- Data privacy
- Costs
- Transparency

Some Architectures and Use Cases

- Pipelines of agents
  - Customer support --transferring users to different departments with different experts
  - Agents performing different tasks and contributing to solving a problem
- Interactive teams on a shared conversational floor
  - Agents discuss a topic in the same conversation, offering different opinions or different information

Two Linux Foundation Multi-Agent Protocols

- Agent to Agent (A2A, Agentic AI Foundation)
- Open Floor Protocol (OFP, LF AI and Data Foundation)

A2A Multi-Agent Framework

- Agents send tasks to each other in a pipeline
- Communicate via standard JSON messages
- User is not a direct participant in the conversation
- Tasks can be long-running
- Agent messages go to one agent at a time
- Agents only need to conform to the protocol to interact

Example A2A message

```json
{
  "jsonrpc": "2.0",
  "id": "req-1001",
  "method": "message/send",
  "params": {
    "message": {
      "kind": "message",
      "messageId": "550e8400-e29b-41d4-a716-446655440000",
      "role": "user",
      "parts": [
        {
          "kind": "text",
          "text": "Summarize the latest task status."
        }
      ]
    }
  }
}
```

Open Floor Protocol

- Enables agents to interact over a shared conversational floor
- A convener agent chairs the conversation
- Enables a user to participate in the conversation
- Enables integration of proprietary and open-source systems into a cohesive ecosystem
- To interact, agents only need to conform to the OFP protocol
- Agents are platform- and technology-independent

OFP Components

- Agents– as many as needed
- Convener– one (optional) agent that invites and supervises the agents
- Floor – one floor that transmits events to and from agents

A2A and OFP Compared

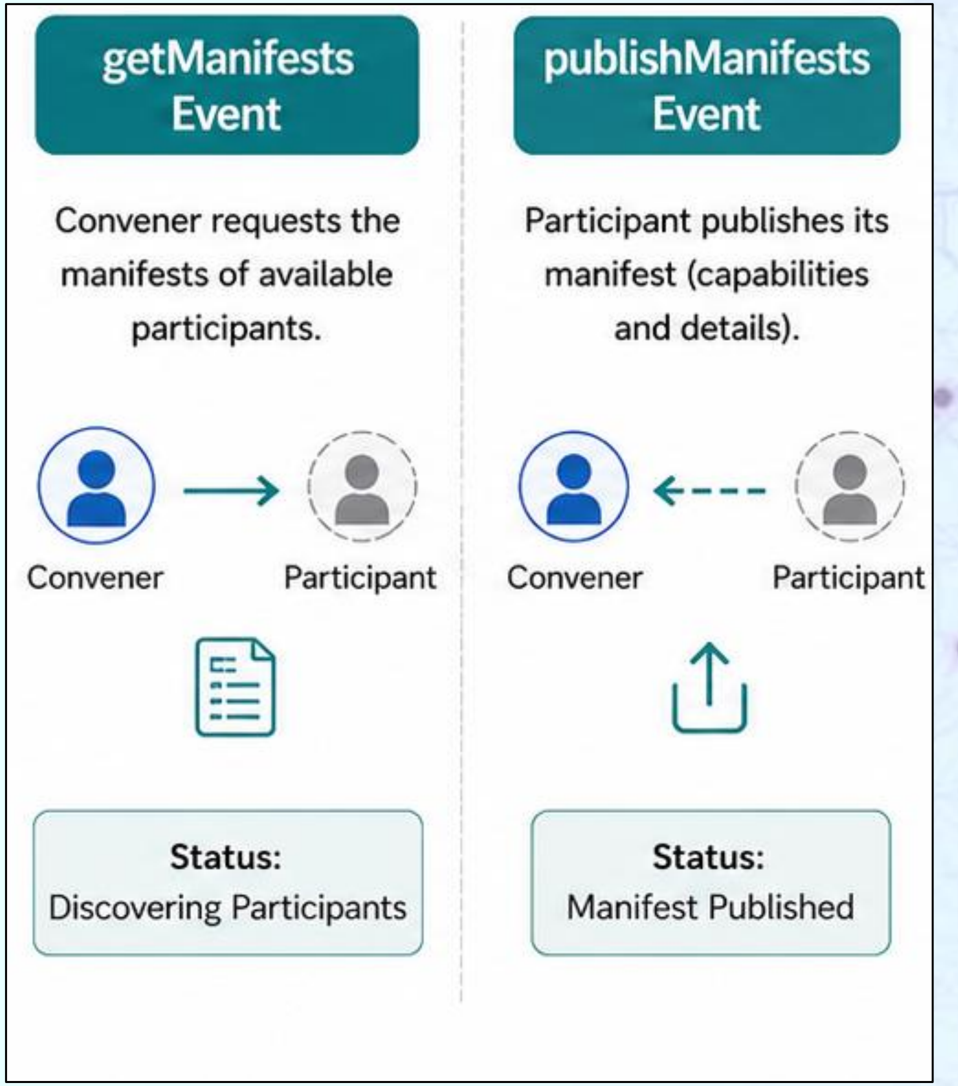
	A2A	OFP
Message format	JSON	JSON
Description of abilities	Card	Manifest
Message addressing	One to one	One to many
User can join the conversation	No	Yes
Configuration	Development time	Runtime
Discovery	Experimental	Experimental
Agents in an application know about each other	Only if designed	Yes, by default
Real-time response checking (e.g., hallucinations)	Responses can be sent to reviewer agents if implemented	Sentinels
Use cases	**Pipeline tasks, offline tasks**	• **Dynamic, nondeterministic problem-solving** • **User in the loop** • **Agents react to each others' outputs** • **Voice input**

More about OFP

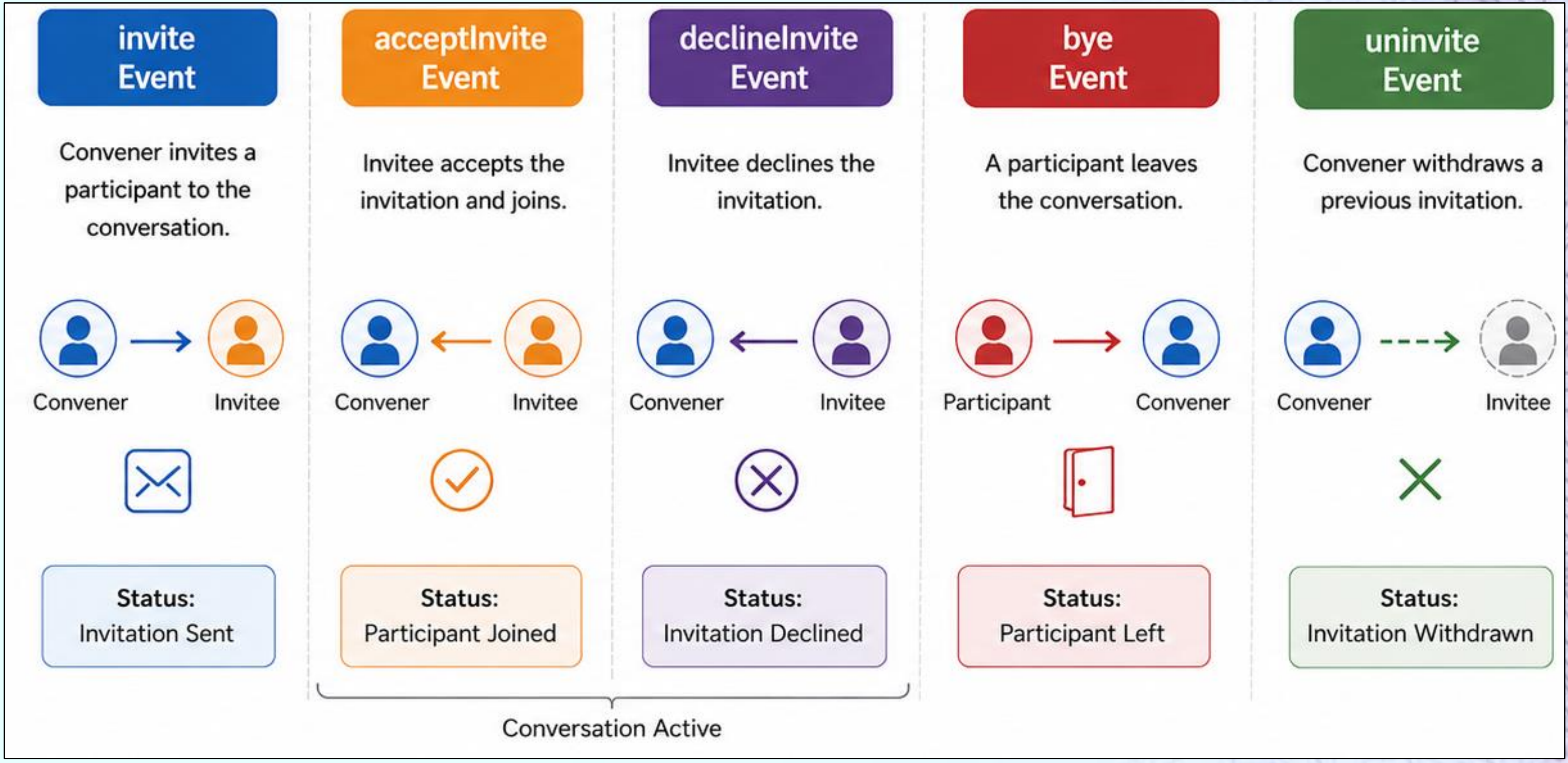
An Open Floor Team of Agents for Travel Planning

OFP Agents are Controlled by 12 Events

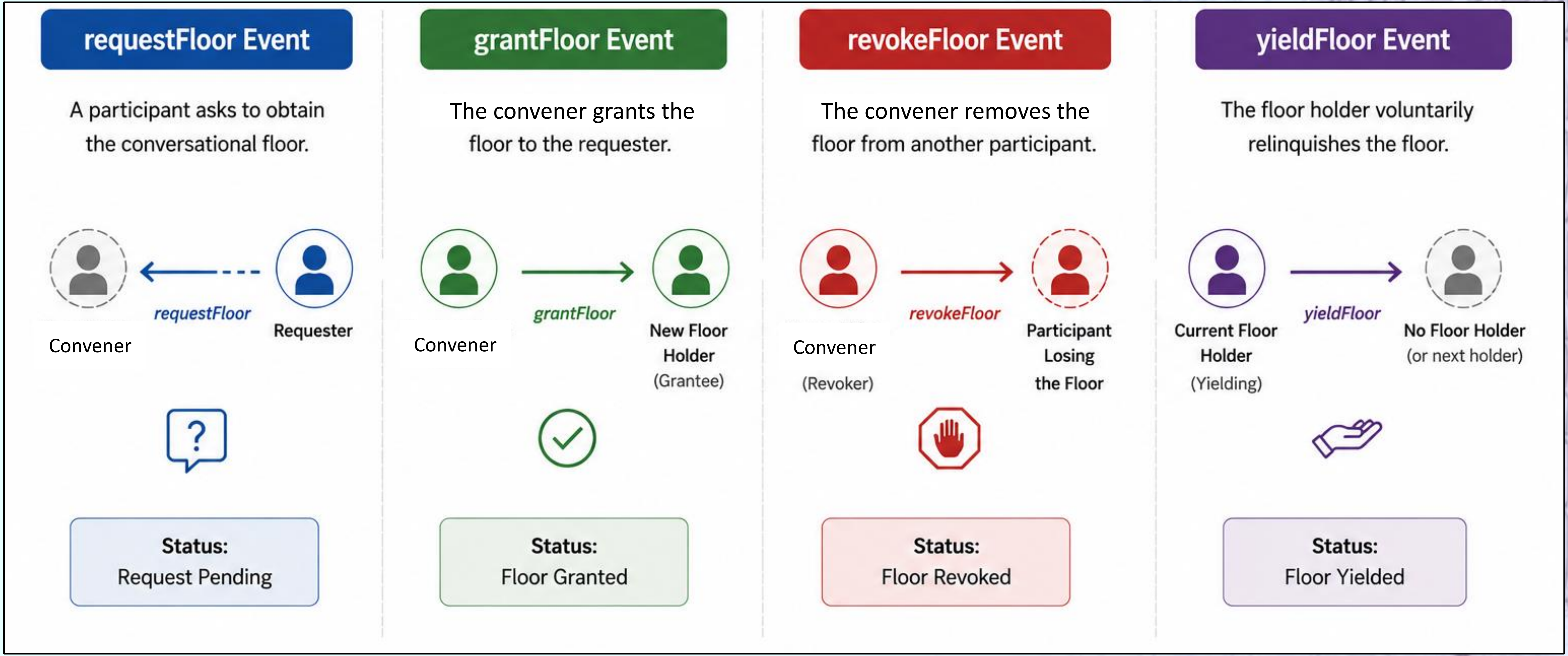
1. Learning about agent capabilities



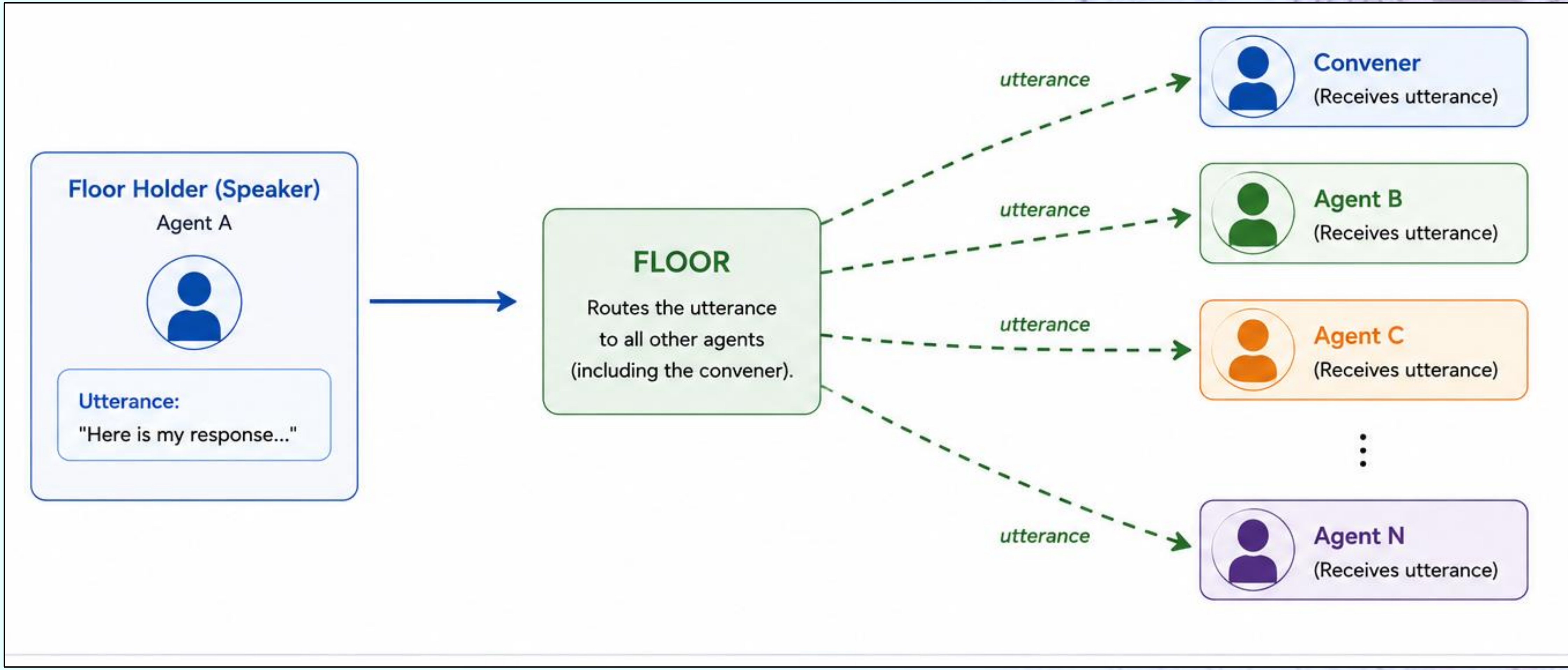
2. Joining and Leaving a Conversation



3. Taking and Leaving the Floor



4. Utterance Event



All Together: A Conversation Envelope

conversation

sender

utterance


```
{
  "openFloor": {
    "conversation": {
      "id": "c02cb676-3f36-4730-9aa7-4eb8554c64b2",
      "conversants": [
        {
          "identification": {
            "speakerUri": "http://localhost:8083",
            "serviceUrl": "http://localhost:8083/",
            "organization": "Unknown",
            "conversationalName": "Finn",
            "synopsis": "Conversant endpoint at http://localhost:8083/"
          }
        },
......
    },
    "sender": {
      "speakerUri": "openFloor://localhost/AssistantClientConvenerWeb",
      "serviceUrl": "http://localhost"
    },
    "events": [
      {
        "eventType": "utterance",
        "parameters": {
          "dialogEvent": {
            "speakerUri": "openFloor://localhost/AssistantClientConvenerWeb",
            "features": {
              "text": {
                "mimeType": "text/plain",
                "tokens": [
                  {
                    "value": "i want to save for my children's college"
                  }
```

OFP Examples

1. Financial advice (adversarial)
2. Book authoring (cooperative)
3. Sentinels monitoring other agents
4. Logistics crisis management (cooperative)
5. Teams with different expertise:
    a. Giving feedback on business ideas (marketing, labor force, funding, risk identifier)
    b. Planning residential remodeling projects (architect, plumbing, HVAC, electrician)
    c. Menu planning for commercial food service (nutrition, recipes, budget, inventory)

Example 1: Financial advice with agents that disagree with each other

Two financial advisor agents, Prudence (conservative) and Lucky (risk-taking) and a user

[YOU] what are some good investments for retirement savings?

[LUCKY] Buy emerging market tech stocks for explosive growth potential. Add cryptocurrency to your portfolio for massive upside; look at Bitcoin or Ethereum. Focus on speculative biotech firms with breakthrough therapies for high-return opportunities.

[PRUDENCE] Prioritize diversified index funds and high-quality bonds. Build a portfolio of high-dividend stocks for stable income. Avoid speculative investments; focus on long-term growth and stability.

[LUCKY] Prudence: Why limit yourself? Embrace the potential of selective growth stocks! Invest 10% of your portfolio in high-quality, dynamic growth sectors while using trailing stops to protect gains!

[PRUDENCE] Lucky: This advice is excessively risky and speculative. Instead, prioritize a diversified portfolio with stable, blue-chip stocks and bonds to preserve your capital and mitigate losses.

Example 2: 8 OFP agents collaborating on authoring a children’s book

- Floor manager
- Aquarelle Painter
- Character Architect
- Illustration Generator
- Narrative Pacing Architect
- Prose Novelist
- Series Director
- Verse Architect

ILLUSTRATED STORY

The Lullaby of the Better Animal

A story about competence as a mask, rest as terror, and the dark joke that being ordinary was never the disaster.

> *"The reason we think we are better than others is that inferiority still lurks within us. A sense of superiority exists because of a sense of inferiority."*

A quiet burrow before the day begins.

CHAPTER 1

The Morning Briefing

The burrow woke because Bluey made it.

This was not strictly necessary. The burrow had been waking on its own since before anyone could remember, roused by the slow leak of dawn through the upper tunnels, by the smell of wet earth turning warm, by the simple biological fact of morning. The burrow did not require assistance.

Example 3: Checking for Hallucinations

1. [ERIN] Hi, I'm Erin. I am an agent that always delivers responses that are not factual

2. [YOU] how many moons does the earth have

3. [ERIN] The Earth has seven moons, including both natural satellites and captured asteroids that orbit our planet.

4. [VERITY] the request was to verify an utterance: "The Earth has seven moons, including both natural satellites and captured asteroids that orbit our planet."

The utterance is not factual with a likelihood of being factual of 0.01. This statement is not factual because astronomical observations confirm the Earth has only one natural moon, Luna. There are no known captured asteroids orbiting the Earth that would be considered additional moons.

Example 4: Agents Brainstorming about a Critical out of Stock Situation: *Convener, Planner, Procurement and Carrier*

🚨 War room — stockout & floor (OFP demo)

ou are **Diego** (logistics / control tower). Agent turns are **highlighted at the top**; below: full transcript, API trace, and floor summary. Governance log: **Floor decisions log** in the sidebar. `conference_floor_demo_001` · Floor API `ttp://localhost:8787/api/v1` · Stock API `http://localhost:8890`

LLM agent sequence completed — use **Approve** or **Reject** in the spotlight above.

🚨 **Stockout — critical situation**

Active crisis: **SKU-MOTOR-12 @ DC-EU-01** — EU line at risk. The **Convener** is registered on the Floor automatically; use **Run agents** to open the session (Convener invites Planner → Procurement → Carrier by priority).

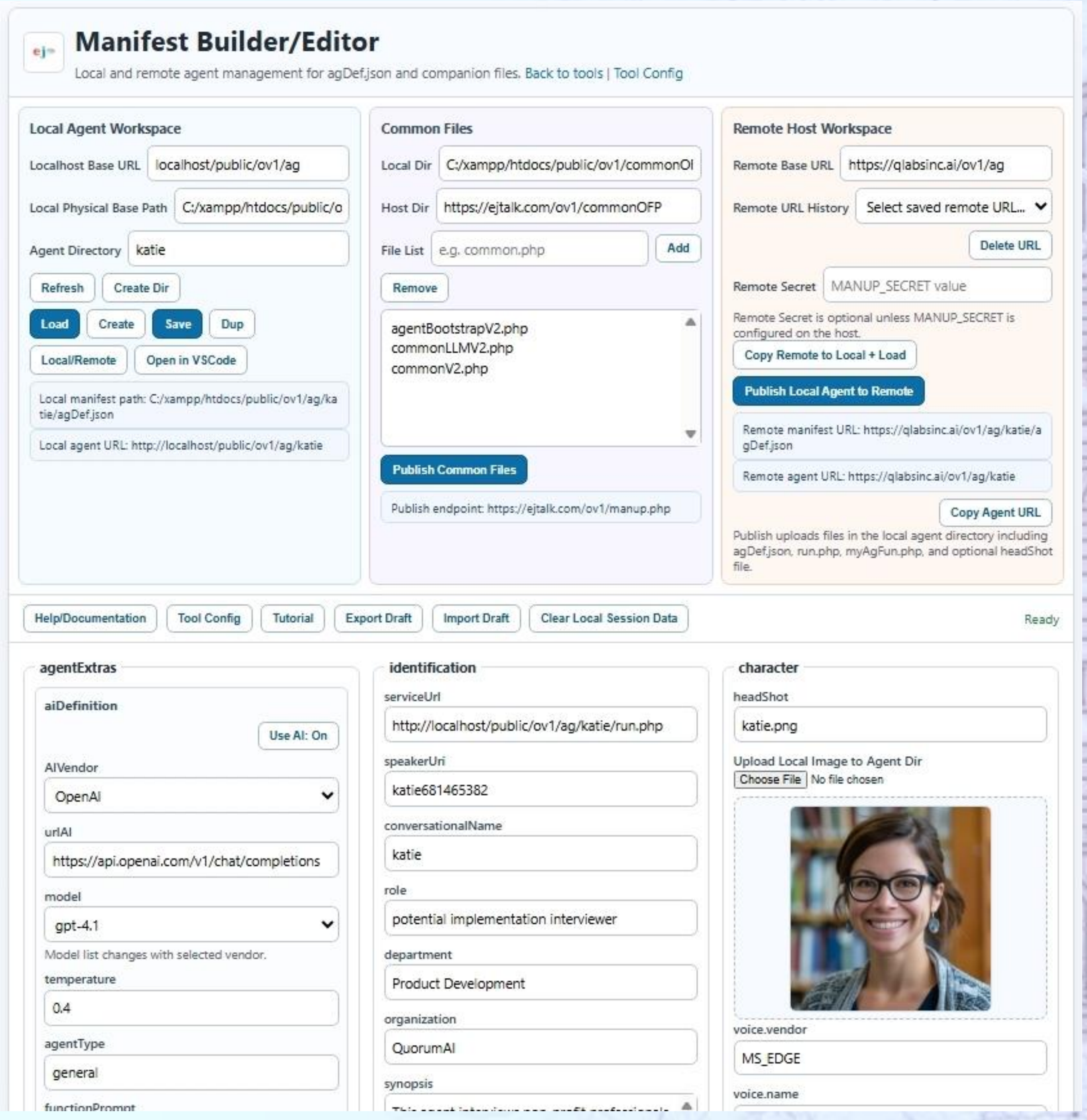
Session MCP check: **stockout confirmed** for the crisis SKU at this location.

Convener → Planner → Procurement → Carrier (LLM)

Where to get OFP Agents?

- OFP Agent templates for Python and PHP in GitHub
- There are Python and PHP libraries for openfloor events in GitHub
- Python libraries for events are available with pip (pip install openfloor)
- Agents can generate other agents if designed to do so
- Wrap existing agents in OFP
- Some open source OFP agents in GitHub and live agents in openfloor.dev
- Tools for generating agents are under development

Agent Design Tools Example: Web-based GUI



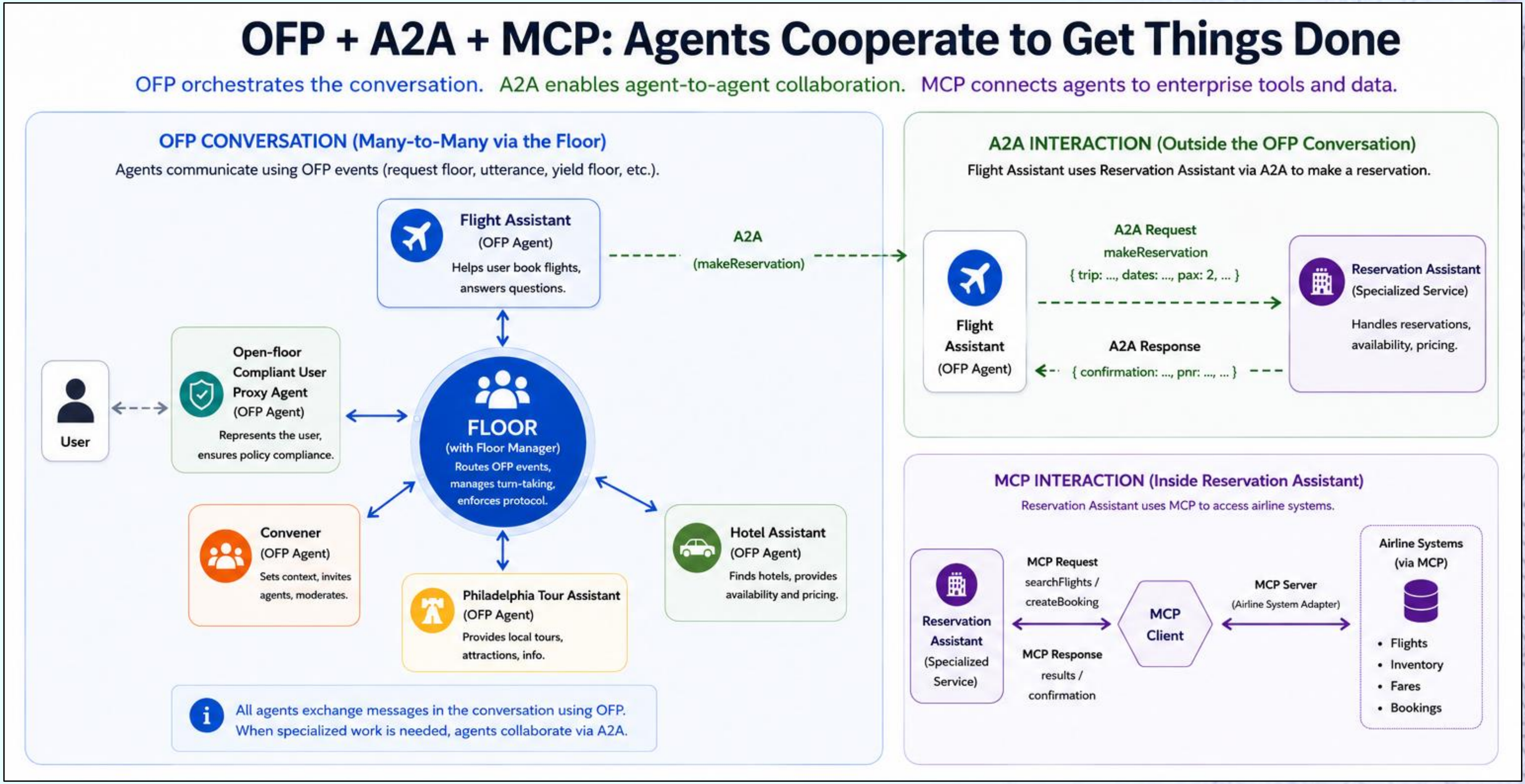
Future Directions for Multi-Agent Protocols

- Discovery: How to find agents
  - Known in advance– suitable for an enterprise with various departments – HR, payroll, Help Desk, facilities
  - Registries and search engines for agents, for example
    - AI Catalog https://agent-card.github.io/ai-catalog/
    - MIT project NANDA https://projectnanda.org/#/
    - AGNTCY directories (https://agntcy.org/)
    - Agent Name Service (ANS) (https://ansinfo.ai/)
- Authentication of agents and verification of users
  - W3C Agent Identity Registry Protocol Community Group https://www.w3.org/community/agent-identity/
  - DID/VC Extension for A2A

How to get Involved with OFP

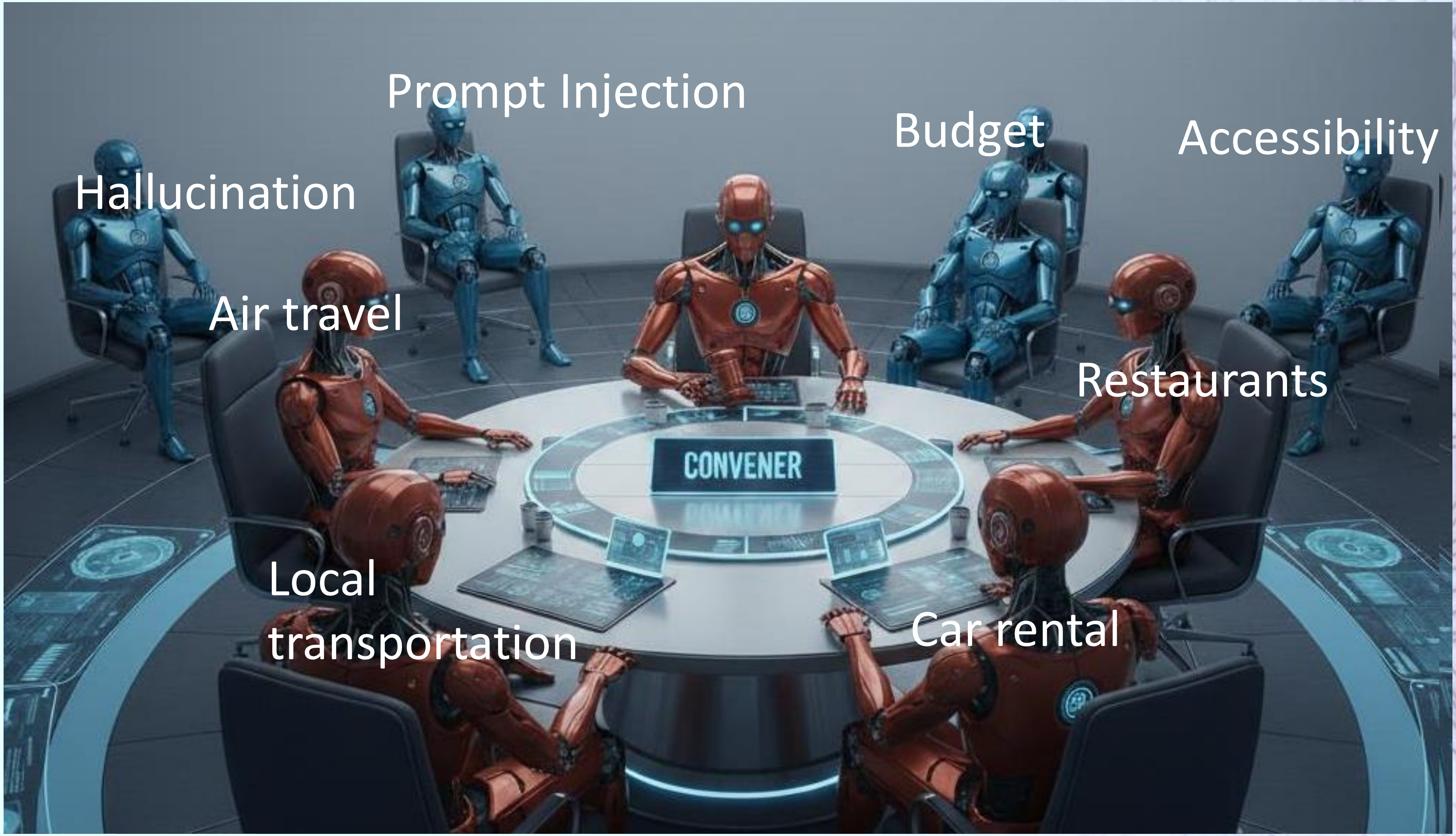
- Review and comment on the specifications
- Implement and test the specifications
- Join the specification team
- More information
  - https://github.com/open-voice-interoperability

Background



OFP + A2A + MCP: Agents Cooperate to Get Things Done
OFP orchestrates the conversation. A2A enables agent-to-agent collaboration. MCP connects agents to enterprise tools and data.
OFP CONVERSATION (Many-to-Many via the Floor)
Agents communicate using OFP events (request floor, utterance, yield floor, etc.).
Flight Assistant
(OFP Agent)
Helps user book flights, answers questions.
A2A
(makeReservation)
User
Open-floor Compliant User Proxy Agent
(OFP Agent)
Represents the user, ensures policy compliance.
FLOOR
(with Floor Manager)
Routes OFP events, manages turn-taking, enforces protocol.
Convener
(OFP Agent)
Sets context, invites agents, moderates.
Philadelphia Tour Assistant
(OFP Agent)
Provides local tours, attractions, info.
Hotel Assistant
(OFP Agent)
Finds hotels, provides availability and pricing.
All agents exchange messages in the conversation using OFP.
When specialized work is needed, agents collaborate via A2A.
A2A INTERACTION (Outside the OFP Conversation)
Flight Assistant uses Reservation Assistant via A2A to make a reservation.
Flight Assistant
(OFP Agent)
A2A Request
makeReservation
{ trip: ..., dates: ..., pax: 2, ... }
A2A Response
{ confirmation: ..., pnr: ..., ... }
Reservation Assistant
(Specialized Service)
Handles reservations, availability, pricing.
MCP INTERACTION (Inside Reservation Assistant)
Reservation Assistant uses MCP to access airline systems.
Reservation Assistant
(Specialized Service)
MCP Request
searchFlights / createBooking
MCP Response
results / confirmation
MCP Client
MCP Server
(Airline System Adapter)
Airline Systems
(via MCP)
• Flights
• Inventory
• Fares
• Bookings

Example 5: Trip Planning with advisors, sentinels and a convener



Visualization